

- Q.28 What are different methods of improving duty ?
- Q.29 Write different methods of groundwater recharge.
- Q.30 What are different types of centrifugal pumps ?
- Q.31 Describe furrow method of irrigation along with its adaptability.
- Q.32 What are different characteristics of aquifer influencing yield of well
- Q.33 Define Infiltration rate and storage coefficient
- Q.34 Describe various factors affecting frequency of irrigation
- Q.35 Explain the volumetric method of water measurement

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Describe the operation and maintenance of centrifugal pump, their common trouble and remedies.
- Q.37 Explain the sprinkler irrigation system in detail. Also describe its adaptability and list the components with layout diagram.
- Q.38 List different surface methods of irrigation. Explain the basic details, characteristics, type and adaptability of furrow irrigation method.

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Roll No.

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3rd Sem / Branch : AGRI ENGG.

Subject:- Farm Irrigation Engg. / Water Res. Engg.

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The Sprinkler irrigation method is recommended in
- Highly undulating area
 - Soils having infiltration rate
 - Irrigation source is a tubewell
 - In all the above situations
- Q.2 The actual area irrigated in a year from outlet is called
- Irrigation period
 - Intensity of Irrigation
 - Irrigation frequency
 - None of these
- Q.3 The total depth of water application to the crop is called :
- Duty
 - Delta
 - Base
 - None of these
- Q.4 The number of days that can be allowed for applying one irrigation to a given design area during the peak consumptive use period of crop being irrigated
- Duty
 - Irrigation period
 - Irrigation frequency
 - Delta

- Q.5 The maximum efficiency of a centrifugal pump is
 a) 50% b) 70%
 c) 85% d) 100%
- Q.6 The downward movement of water through the soil is the
 a) Infiltration b) Absorption
 c) Porosity d) None of the above
- Q.7 The ratio of the quantity of water delivered to the field and quantity of water pumped into the canal
 a) Water storage efficiency
 b) Water application efficiency
 c) Water conveyance efficiency
 d) Water use efficiency
- Q.8 The line connecting points having the same depth below a water surface.
 a) Isobar b) Isobath
 c) Contour d) Drawdown
- Q.9 The total amount of water applied through irrigation is termed as
 a) Net irrigation requirement
 b) Gross irrigation requirement
 c) Duty
 d) Delta
- Q.10 The height at which water stand in a piezometer
 a) Static water level b) Piezometric surface
 c) Pumping water level d) None of these

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 List two factors affecting frequency of irrigation.
 Q.12 State two benefits of sprinkler irrigation.
 Q.13 What is the purpose of Canal lining ?
 Q.14 List two physical properties of soil.
 Q.15 What is the use of priming ?
 Q.16 Name two method to improve duty
 Q.17 What are different sources of water for irrigation
 Q.18 State two units of water measurement
 Q.19 List different types of impellers
 Q.20 Define water application efficiency

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the criteria for selection of irrigation pumps
 Q.22 What are ill effects of irrigation ?
 Q.23 Describe the working of submersible pump
 Q.24 What are different types of impellers used in centrifugal pump ?
 Q.25 Explain any one method of measurement of water in open channels
 Q.26 List different types of canal lining ? Write the advantages of canal lining.
 Q.27 What is the criteria for selection of irrigation pumps?

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